Twin-Pulse

ML-Based Digital Twin Framework for AI-Driven Asset Prognosis

Predict before failure. Perform without interruption.

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1 Executive Summary

Modern organisations across manufacturing, energy, utilities, transportation, and even healthcare depend on critical equipment that must operate reliably. Unexpected equipment failure leads to production losses, service disruption, emergency repair costs, and reputational risk. Although most assets today are equipped with sensors, many organisations still rely on fixed maintenance schedules or simple alarm thresholds. These methods do not capture gradual degradation and often fail to provide sufficient early warning before breakdown.

This project addresses a strategic business question:

> How can organisations move from reactive maintenance to proactive, data-driven decision-making using sensor data?

The proposed solution is a Machine Learning–driven Digital Twin framework — a virtual representation of a physical asset that continuously monitors condition, detects degradation, and estimates failure risk.

While the framework is general and applicable to multiple industries, this study demonstrates the approach using industrial pumps as a case example. Pumps are widely used across sectors and represent a realistic, high-impact maintenance challenge.

The system performs three core functions: it classifies equipment into four health stages (Healthy, Degrading, Critical, and Failure), converts these predictions into a 0–100 risk score for managerial interpretation, and estimates the remaining days before likely failure to support planning decisions. Under structured simulation conditions, the framework achieved 91% accuracy in health stage identification, 89% recall in detecting failure events, and an average error of 1.79 days in time-to-failure prediction, while consistently generating early-warning signals within a 5–30 day decision window.

The primary target audience includes operations managers, maintenance heads, plant managers, asset reliability teams, and strategic decision-makers overseeing equipment-intensive operations. From a managerial perspective, the value is directly actionable: the system enables prioritisation of high-risk assets across a fleet, supports planned maintenance instead of emergency intervention, improves spare part procurement timing, reduces downtime, enhances asset availability, and facilitates data-backed maintenance budgeting.

Instead of responding after a breakdown, organisations can intervene at the Degrading or Critical stage, when corrective action is less disruptive and less expensive.

In conclusion, this project demonstrates that integrating machine learning with a Digital Twin framework enables organisations to shift from reactive maintenance to proactive, risk-based asset management — improving operational resilience and strategic control.

2 Problem Statement & Business Context

2.1 Industry / Domain

This project operates in the domain of predictive maintenance and asset intelligence, applicable across manufacturing, energy, utilities, transportation, infrastructure, and healthcare equipment management.

Industrial pumps are used as a representative case study, but the framework is generalisable to any sensor-equipped critical asset.

2.2 Real-World Context

Organisations today generate continuous equipment data through vibration, temperature, pressure, power, and acoustic sensors. Despite this, most maintenance strategies remain:

- Reactive (repair after breakdown)
- Alarm-based (threshold triggers)
- Time-based (scheduled servicing regardless of condition)

These approaches do not model gradual degradation. As a result, failures are often detected too late, or maintenance is performed too early — both leading to financial inefficiency.

The key gap is converting sensor data into structured decision intelligence.

2.3 Target Stakeholder

Primary stakeholders include operations and plant managers, maintenance heads, reliability engineers, asset management teams, and strategic operations leaders. These stakeholders require early-warning signals, risk prioritisation across multiple assets, time-to-failure estimates, and scenario-based planning tools rather than raw technical outputs.

2.4 Decision Being Optimised

The central decision being optimised is when and how maintenance should be performed to minimise downtime while maximising asset life and operational continuity. This involves identifying which asset requires attention first, determining the urgency of intervention, estimating the remaining time before failure, and evaluating whether intervention at the present time meaningfully alters future risk.

2.5 Nature of the Analytical Problem

The problem is structured as a layered predictive pipeline comprising health stage classification (categorical), risk quantification (0–100 managerial index), Remaining Useful Life (RUL) regression, scenario simulation through the Digital Twin Sandbox, and AI-based diagnostic interpretation via the Agentic layer. These components operate sequentially to transform raw sensor data into actionable maintenance intelligence, moving beyond isolated prediction toward an integrated Digital Twin architecture.

2.6 Why Machine Learning is Appropriate

Machine learning is appropriate because equipment degradation is non-linear and multi-sensor dependent, static thresholds cannot capture rate-of-change or trend acceleration, different machines degrade differently, and early-warning signals emerge from temporal patterns rather than single values. Traditional rule-based systems answer “Has the threshold been crossed?” whereas this framework answers “How fast is degradation accelerating, how risky is the asset, how many days remain, and what happens if we intervene now?” By combining ML prediction, deterministic risk modelling, Digital Twin simulation, and AI-driven explanation, the system transforms raw sensor data into structured decision intelligence.

3 Data Understanding & Variable Design

3.1 Data Source

The project uses a physics-informed synthetic dataset designed to realistically simulate industrial equipment degradation. While the framework is industry-agnostic, the dataset models 15 industrial pumps observed hourly over six months (180 days), generating approximately 65,000 observations with eight sensor streams per asset. The health lifecycle follows a progressive pattern (Healthy $\rightarrow$ Degrading $\rightarrow$ Critical $\rightarrow$ Failure), with 12 pumps reaching failure and 3 censored within the observation window.

A synthetic dataset was selected because real industrial failure data is often confidential and limited. The data generation process incorporates non-linear degradation, sensor interdependencies, operational noise and drift, transient disturbances, and overlapping health ranges to ensure realistic modelling complexity.

3.2 Variable Categorization

3.2.1 Dependent Variables

The dependent variables include Health Stage (a categorical classification target with four states: Healthy, Degrading, Critical, and Failure) and Remaining Useful Life (a numerical regression target representing days until failure).

3.2.2 Independent Variables

The eight numerical sensor variables are time-series numerical streams:

Radial vibration	Axial vibration
High-frequency vibration	Bearing temperature
Casing temperature	Discharge pressure
Power consumption	Acoustic emission

3.2.3 Other Variables

Other variables include Machine ID (categorical identifier), Timestamp (time index variable), and Lifecycle Position (elapsed time). These are not predictive features directly but are essential for temporal grouping and leakage prevention.

3.3 Feature Engineering

Raw sensor values alone are insufficient to detect gradual degradation; therefore, time-aware transformations were applied. Rolling statistics (48h and 168h windows), including rolling mean, standard deviation, minimum, maximum, and range, were used to capture short-term versus long-term stability patterns. A 24-hour delta was computed to detect abrupt behavioural shifts. Linear regression slopes (48h and 168h) were included to quantify rate-of-change, since degradation velocity is often more informative than absolute magnitude. Lifecycle position was incorporated into the RUL model to provide temporal context for failure timing estimation.

No PCA or dimensionality reduction was applied, as interpretability was prioritised over compression. Rows without sufficient historical context were removed to preserve temporal causality. The final dataset contained 136 engineered features per observation.

3.4 Exploratory Data Analysis (EDA) Insights

EDA revealed progressive sensor escalation, with vibration and temperature signals increasing non-linearly as failure approached, indicating that early degradation manifests as trend acceleration rather than threshold breaches. Sensor ranges overlapped across health stages, confirming that simple rule-based limits are insufficient. Rate-of-change features provided stronger discrimination between Degrading and Critical states than static magnitudes, reinforcing the importance of monitoring acceleration. Transient outliers were observed but did not consistently indicate failure, highlighting the need to distinguish temporary anomalies from sustained degradation.

3.5 Why Preprocessing Was Necessary

Preprocessing ensured temporal causality, prevented cross-machine leakage, captured degradation velocity, enhanced early-warning sensitivity, and maintained managerial interpretability. The objective was not feature complexity, but degradation intelligence.

4 Model Selection & Justification

The modelling framework consists of health stage classification (multi-class) and Remaining Useful Life (RUL) estimation (regression).

4.1 Algorithms Considered

For classification, Logistic Regression, Decision Trees, Random Forest, LightGBM, and a Soft-Voting Ensemble were considered. For RUL regression, Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor were evaluated. Neural Networks and SVMs were not prioritised due to dataset size, interpretability requirements, and overfitting risk.

4.2 Final Model Selection & Justification

4.2.1 Health Classification – Soft-Voting Ensemble

The dependent variable is categorical with non-linear sensor interactions. Tree-based models were preferred to capture multi-sensor dependencies: Random Forest reduces variance through averaging, LightGBM reduces bias via sequential error correction, and Logistic Regression improves probability calibration. The ensemble balances bias–variance tradeoffs and improves robustness under class imbalance, prioritising architectural stability over marginal accuracy gains.

4.2.2 RUL Estimation – Gradient Boosting Regressor

RUL is continuous and influenced by non-linear degradation velocity. Linear regression was insufficient, while Gradient Boosting was selected for its ability to model complex tabular relationships, iteratively minimise error, perform reliably with moderate dataset size, and maintain controlled variance through shallow trees.

4.3 Mathematical Intuition (High-Level)

Logistic Regression minimises cross-entropy loss to optimise class probabilities. Random Forest builds multiple bootstrapped trees and averages predictions to reduce variance. Gradient Boosting iteratively minimises prediction error (e.g., MSE) by fitting residuals. Soft Voting aggregates predicted class probabilities and selects the highest combined probability. Model selection was guided by the nature of target variables, non-linear degradation behaviour, interpretability requirements, bias–variance balance, and deployment feasibility.

5 Model Evaluation & Comparison

Model evaluation was conducted separately for health stage classification and Remaining Useful Life (RUL) estimation. Given class imbalance and operational risk implications, multiple complementary metrics were used rather than relying solely on accuracy.

5.1 Evaluation Metrics & Justification

Accuracy measures overall prediction correctness but is insufficient in imbalanced datasets, as a model could achieve high accuracy by predominantly predicting the majority class. Precision evaluates how many predicted failures are correct, while recall measures how many actual failures are detected; recall is particularly critical in maintenance contexts where missed failures are costly. The confusion matrix provides class-wise breakdowns, revealing operationally meaningful misclassifications such as confusion between Degrading and Critical stages. ROC–AUC measures class separability independent of a fixed threshold, making it suitable for probability-based decision systems.

For RUL regression, Mean Absolute Error (MAE) reflects average forecasting deviation in days and is directly interpretable, while Root Mean Squared Error (RMSE) penalizes larger deviations and highlights sensitivity to outliers. Machine-level stratified cross-validation was applied to prevent data leakage and ensure that evaluation reflects realistic deployment scenarios where entire machines remain unseen during training.

5.2 Classification Model Performance

Table 1: Classification Performance Metrics

Metric	Value	Interpretation
Accuracy	91.08%	Strong overall classification across four health stages
Macro F1 Score	0.874	Balanced performance across minority and majority classes
Failure Recall	89%	High sensitivity to rare failure events
Macro Precision	0.857	Moderate conservatism in early-stage detection
ROC-AUC (Multi-class)	High (threshold-independent)	Strong class separability

Interpretation:
The model reliably detects degradation and failure stages while maintaining balanced performance under class imbalance. High recall reduces risk of missed failures.

5.3 RUL Regression Model Performance

Table 2: RUL Regression Performance Metrics

Metric	Value	Interpretation
MAE	1.79 days	High precision within operational planning window
RMSE	3.75 days	Larger errors are limited and infrequent
$\pm$7 Day Accuracy	97.48%	Strong reliability within practical decision horizon
Mean Bias	+1.73 days	Slight conservative overestimation of failure time

Interpretation:
The RUL model provides operationally actionable forecasts within the 5–30 day planning window. Slight positive bias indicates conservative prediction, which is safer in maintenance planning.

5.4 Overall Evaluation

The integrated system demonstrates strong early degradation detection, reliable failure sensitivity, accurate time-to-failure estimation, and stable performance under simulated non-linear degradation. Performance metrics confirm that the layered Digital Twin architecture is both statistically robust and operationally practical.

6 Technology Stack & Architecture

6.1 Technology Stack

The project was implemented in Python due to its strong ecosystem for machine learning and rapid prototyping. Core libraries included pandas and numpy for data processing, scikit-learn and LightGBM for modelling, imbalanced-learn (SMOTE) for class imbalance handling, joblib for model serialization, and Streamlit with matplotlib for dashboard visualisation. Deep learning frameworks were not used, as the dataset size and structured tabular format did not require neural architectures.

6.2 Architecture Overview

The system follows a layered Digital Twin architecture: Sensor Data $\rightarrow$ Feature Engineering $\rightarrow$ Health Classification $\rightarrow$ Risk Scoring $\rightarrow$ RUL Estimation $\rightarrow$ Dashboard and Sandbox $\rightarrow$ AI Narrative. A streaming simulator processes data sequentially to emulate real-time deployment, while the Digital Twin Sandbox enables scenario testing by comparing baseline and intervention trajectories. An optional Agentic AI layer converts model outputs into structured maintenance insights for decision-makers.

6.3 Dashboard Interface and User Experience

The final product is delivered through a multi-page Streamlit dashboard designed across three operational layers.

At the fleet level, the Fleet Overview page enables managers to monitor all assets simultaneously using health stage indicators, risk bands, and prioritisation tables, allowing rapid identification of high-risk machines across the portfolio.

At the asset level, the Detailed View and Sensor Trend pages provide machine-specific analysis, including historical sensor behaviour, degradation velocity, risk progression, and Remaining Useful Life estimates. This layer supports maintenance engineers in diagnosing underlying deterioration patterns.

At the strategic decision layer, the Digital Twin Sandbox enables users to simulate maintenance interventions and compare baseline versus intervention trajectories in terms of risk evolution and predicted failure timing. The sandbox dynamically recalculates health stage transitions, risk scores, and Remaining Useful Life under both intervention and no-intervention scenarios, enabling quantitative comparison of maintenance timing decisions.

The AI Diagnostic Agent synthesizes classification probabilities, risk scores, and RUL predictions into structured maintenance recommendations. It generates prioritised intervention schedules, highlights urgency levels, and explicitly communicates projected consequences if intervention is delayed, ensuring interpretability and decision support for non-technical stakeholders.

Together, these layers transform predictive outputs into operational intelligence, making the system usable across fleet managers, maintenance engineers, and strategic planners.

6.4 Deployment and Limitations

Models are serialized using joblib, and the current implementation simulates real-time inference while remaining adaptable for integration with live sensor feeds. Key limitations include validation on synthetic data, absence of cloud-based deployment, use of point-estimate RUL without uncertainty intervals, and replay-based sandbox simulation rather than forward probabilistic forecasting. Despite these constraints, the architecture remains modular and extensible.

7 Business Impact & Recommendations

7.1 Business Value Creation

If implemented in an industrial setting, the proposed Digital Twin framework enables a shift from reactive to condition-based maintenance, generating measurable operational and financial value. Downtime reduction is a primary driver, as early warning within a 5–30 day window allows maintenance to be scheduled during planned shutdowns; even a 10–20% reduction in unplanned downtime (estimated based on industry benchmarks) can yield significant savings in equipment-intensive environments. Maintenance cost optimization is achieved by replacing time-based servicing with condition-based intervention, potentially reducing unnecessary part replacement, overtime labour, and emergency repair premiums, with estimated savings of 5–15% depending on asset criticality.

Risk mitigation is strengthened through high failure recall (89%), reducing the probability of catastrophic breakdown and improving operational reliability, safety compliance, and asset life management. Accurate Remaining Useful Life estimates (MAE = 1.79 days) further support capital planning by improving spare part procurement timing, workforce allocation, and long-term asset replacement strategy. The framework is also adaptable beyond industrial pumps to manufacturing systems, power infrastructure, healthcare equipment, and other sensor-equipped assets, extending its strategic applicability.

7.2 Managerial Recommendations

Organisations should transition from time-based to risk-based maintenance scheduling, use structured risk bands (Normal, Watch, Warning, Critical) for fleet prioritisation, monitor degradation velocity rather than static thresholds, integrate RUL forecasts into spare part and workforce planning, and pilot the framework on high-criticality assets prior to fleet-wide deployment.

8 Limitations & Future Scope

8.1 Limitations

The study relies on a physics-informed synthetic dataset rather than real industrial telemetry. Although designed to reflect realistic degradation behaviour, synthetic data may not capture the full operational variability of live environments, and the fleet size of 15 assets may limit generalisation to very large-scale deployments. Failure threshold tuning using evaluation data may introduce mild optimism bias, and class imbalance (few failure samples) may influence sensitivity metrics despite SMOTE correction.

While ensemble methods reduce variance, boosting models remain susceptible to overfitting if degradation patterns are overly structured. Although machine-level splitting mitigates leakage, real-world noise may affect stability. From an ethical and operational standpoint, over-reliance on automated predictions may reduce human oversight, incorrect RUL estimates could influence high-stakes decisions, and AI-generated diagnostic narratives require validation before execution. The system is designed as decision support rather than autonomous control.

Scalability remains constrained, as the current implementation is single-machine simulation-based, lacks cloud-distributed deployment, and does not incorporate real-time IoT integration or high-frequency streaming.

8.2 Future Scope

Future work includes validation on real industrial datasets, probabilistic RUL estimation with uncertainty intervals, advanced ensemble or deep learning methods for larger-scale deployments, Explainable AI techniques (e.g., SHAP) for feature-level interpretability, cloud-based real-time deployment architecture for multi-asset scaling, and forward-projection modelling within the Digital Twin Sandbox instead of replay-based simulation.

While current results are validated under controlled conditions, the architecture remains modular and extensible toward real-world deployment and advanced predictive intelligence.

Disclaimer: Generative Artificial Intelligence (GenAI) tools were utilised solely to enhance the language quality and clarity of this project. All ideas, concepts, analytical decisions, and conclusions are the original contributions of Group 6 and are based on research conducted independently by the group.